Questions

1. Show that

$$\int_0^{\pi/2} \frac{\cos^5\theta}{2+\sin\theta}\,\mathrm{d}\theta = 9\ln\left(\frac{3}{2}\right) - \frac{41}{12}$$

[**8**]

2. **(a)** Use the substitution $u = 1 + x^2$ to show that

$$\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x = \int_p^q \frac{(u-1)^4}{2u}\,\mathrm{d}u$$

where p and q are constants to be found. **[3]**

(b) Hence show that

$$\int_0^2 \frac{x^9}{1+x^2}\,\mathrm{d}x = A + B\ln 5$$

where A and B are constants to be found. **[5]**

3. Show that

$$\int \frac{\sqrt{x^2-16}}{x^2}\,\mathrm{d}x = \operatorname{arcosh}\left(\frac{x}{4}\right) - \frac{\sqrt{x^2-16}}{x} + c$$ **[6]**

4. Find the exact value of

$$\int_0^{\frac{1}{2}\ln(2+\sqrt{3})} \frac{\operatorname{sech}^2 x}{\sqrt{4-9\tanh^2 x}}\,\mathrm{d}x$$

[8]

5. Using calculus, find the exact values of

(i)
$$\int_1^2 \frac{1}{x^2+3x+2}\,\mathrm{d}x$$
[3]

(ii)
$$\int_1^{\sqrt{3}} \frac{\sqrt{4-x^2}}{x^2}\,\mathrm{d}x$$
[5]

6. Find the exact value of

$$\int_2^4 \frac{\sqrt{x^2-4}}{x^2}\,\mathrm{d}x$$

Give your answer in simplest exact form. **[9]**

7. Show that

$$\int \frac{x^2}{\sqrt{x^2+16}}\,\mathrm{d}x = \frac{1}{2}x\sqrt{x^2+16} - 8\,\operatorname{arsinh}\left(\frac{x}{4}\right) + c$$ **[6]**

8. Show that
$$\int_0^{\ln 2} \frac{e^{3x} + 2e^{2x}}{\sqrt{e^x + 4}} \, \mathrm{d}x = \frac{32\sqrt{6} - 30\sqrt{5}}{5}$$ **[7]**

9. Using the substitution $u = \frac{2}{3}\tanh x$, evaluate

$$\int_0^1 \frac{\operatorname{sech}^2 x}{\sqrt{9 - 4\tanh^2 x}}\,\mathrm{d}x$$

[8]

10. Show that

$$\int_0^{\pi/2} \frac{\sin 3\theta}{2+\cos\theta}\,\mathrm{d}\theta = 15\ln\left(\frac{3}{2}\right) - 6$$

[7]

11. **(a)** Use the substitution $u = \sqrt{2x+1}$ to show that

$$\int_0^{15/2} \frac{x}{2+\sqrt{2x+1}}\,\mathrm{d}x = \int_p^q \frac{u(u^2-1)}{2(u+2)}\,\mathrm{d}u$$

where p and q are constants to be found. **[3]**

(b) Hence show that

$$\int_0^{15/2} \frac{x}{2+\sqrt{2x+1}}\,\mathrm{d}x = A - B\ln 2$$

where A and B are constants to be found. **[4]**

12. Use an appropriate substitution to find the exact value of

$$\int_0^{\pi/6} \frac{\cos x \cos 2x}{(1+\sin x)^{3/2}}\,\mathrm{d}x$$

Give your answer in the form $a + b\sqrt{6}$, where a and b are rational numbers to be determined. **[7]**

13. Using calculus, find the exact values of

(i)

$$\int_{-2}^{0} \frac{1}{x^2 + 4x + 8} \, \mathrm{d}x$$

[3]

(ii)

$$\int_{0}^{\sqrt{3}} \frac{x^2}{\sqrt{4 - x^2}} \, \mathrm{d}x$$

[5]

14. Show that

$$\int_1^3 \frac{(x-1)^2}{\sqrt{x+1}}\,\mathrm{d}x = \frac{16}{15}(7-4\sqrt{2})$$ **[7]**

15. Use an appropriate substitution to find the exact value of

$$\int_0^{2\sqrt{2}} \frac{x^2}{\sqrt{16-x^2}}\,\mathrm{d}x$$

Give your answer in the form $a + b\pi$, where a and b are rational numbers. **[9]**